FREIE UNIVERSITÄT BERLIN

AI Speech and language variation & change

A study on manifestation of AI speech in human natural language

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Preliminary

inspired by BSI (2025) and Yakura et al. (2025) which found evidence for GPT influenced human language after the introduction of chatGPT we tried to replicate the Yakura et al. (2025) pipeline of building an AI vocabulary (model preferred lemmata) and compare frequencies of model typical words across pre- and post model introduction human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

Questions / Hypotheses

- do humans adapt to language produced by LLM and incorporate model vocabulary into their own language production?
- H1: we will find higher frequencies of LLM typical vocabulary within human natural language corpora after onset of model introduction.

Methods / Data

our human languaga data consists of raw texts from german bundestag plenary protocols, DIP (2026). the LLM corpus consists of model summaries of a first subset of these texts generated with the prompt (to gemini API) you see in C1.

we first devised model typical lemmata in the AI corpus which are distinct for that corpus using their log odds score calculated over pre/post target, Figure 3.

c1

System prompt: You are a member of german parliament. Prepare a summary of the text provided to present at a local community meeting of your party members. Output in german language, no preamble, no extra information, just the plain text. Wordcount maximal 300 words, containing not more than 5% of the keywords of the text provided and explicitly not just a list of keywords but an entertaining text. You are supposed to interprete freely, including background insights on daily politics. Keep in mind thatthe text will be used as is as keynotes to the talk being held to the locals. Text:

p2

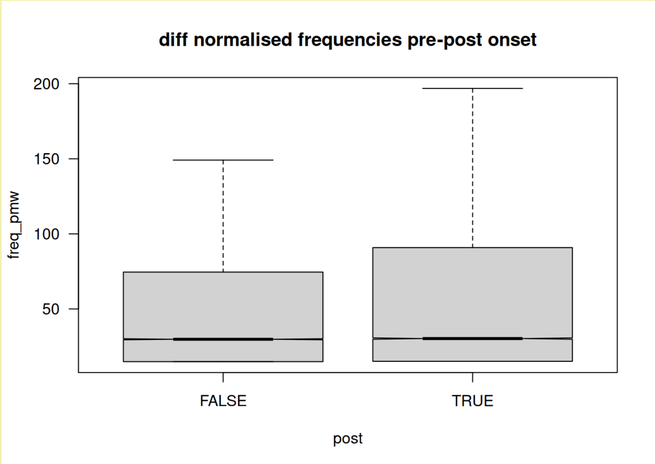

Figure 1: stats descriptive

p3

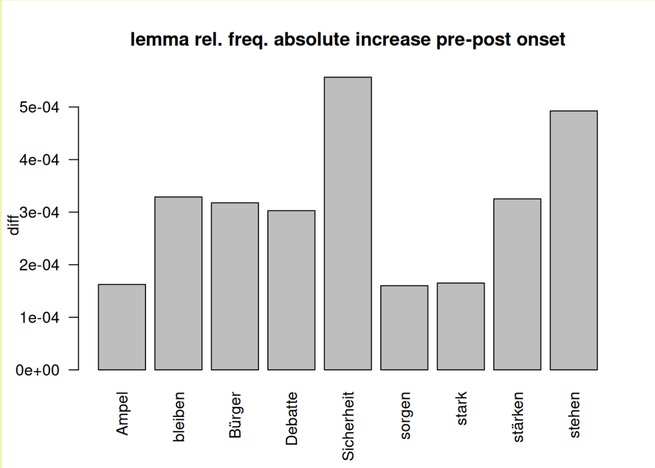

Figure 2: lemma frequency increase

c2: corpus stats

stat	var	pre	post
mean	freq_rel gpt lemmas	0.000704	0.000751
mean	freq normalised pmw	29.825	30.288
tokens	n	1640446	1940385

statistics…

c3: regression simple model

```
Call:
lm(formula = freq_pmw ~ post + gp, data = df_lg.norm)

Residuals:
   Min     1Q Median     3Q    Max
 -1400   -435   -388   -243 436922

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   833.75      38.64  21.576  < 2e-16 ***
postTRUE       83.94      20.64   4.067 4.77e-05 ***
gp             68.11       5.20  13.097  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3765 on 133087 degrees of freedom
  (1393 observations deleted due to missingness)
Multiple R-squared:  0.0014,    Adjusted R-squared:  0.001385
F-statistic: 93.32 on 2 and 133087 DF,  p-value: < 2.2e-16

Analysis of Variance Table

Response: freq_pmw
              Df     Sum Sq    Mean Sq F value    Pr(>F)
post           1 2.1394e+08  213944737  15.096 0.0001022 ***
gp             1 2.4310e+09 2431030842 171.537 < 2.2e-16 ***
Residuals 133087 1.8861e+12   14172027
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

p4

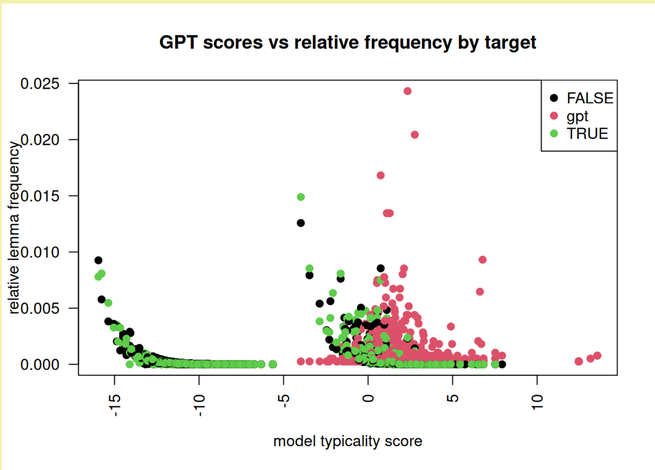

Figure 3: model vocab distribution

Results

to gather an insight, yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, see Figure 1.

the 9 lemma where 1. the relative frequency of lemma in target=post is higher than in target=pre (n=27) and 2. that frequency exceeds the mean relative frequency is displayed in Figure 2

we found significant frequency increases using a linear regression model (Bates et al. (2015)), cf. C3/C4, which tested the hypothesis.

c4: regression mixed effects model

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: freq_pmw ~ post + gp + (1 | lemma)
   Data: df_lg.norm

REML criterion at convergence: 2490672

Scaled residuals:
    Min      1Q  Median      3Q     Max
-48.624  -0.047  -0.032   0.008  74.360

Random effects:
 Groups   Name        Variance Std.Dev.
 lemma    (Intercept) 8839356  2973
 Residual             1754704  1325
Number of obs: 133090, groups:  lemma, 99649

Fixed effects:
              Estimate Std. Error        df t value Pr(>|t|)
(Intercept) 4.344e+02  3.966e+01 1.055e+05  10.953   <2e-16 ***
postTRUE    1.246e+02  9.496e+00 5.602e+04  13.126   <2e-16 ***
gp          2.932e+01  5.723e+00 1.017e+05   5.124    3e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
         (Intr) psTRUE
postTRUE -0.110
gp        0.959  0.009

Type III Analysis of Variance Table with Satterthwaite's method
        Sum Sq   Mean Sq NumDF  DenDF F value    Pr(>F)
post 302319616 302319616     1  56015 172.291 < 2.2e-16 ***
gp    46071208  46071208     1 101716  26.256 2.996e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Limitations

if the findings of increased AI lemma frequencies really prove manifestation of AI language depends also on influences of other factors of language variation and change which Yakura et al. (2025) already took into account. a deeper going analysis of general trends in language development and investigation on the bidirectional influences of model-training and human language could harden the results.

references

Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using Lme4." *Journal of Statistical Software* 67 (1): 1–48. https://doi.org/10.18637/jss.v067.i01.

BSI, Brand Science Institute. 2025. "Wie KI Unsere Sprache Verändert – Eine Empirische Studie." https://www.bsi.ag/cases/104-case-studie-wie-ki-unsere-sprache-veraendert---eine-empirische-studie.html.

DIP. 2026. "DIP - Bundestagsprotokolle." Docs. *DIP - API*. Berlin. https://dip.bundestag.de/%C3%BCber-dip/hilfe/api#content.

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```
get=
s.c),
OP")
SPUN
"wo
ds only
PUND
cept)"]]), ]
$word[i
response <- POST(geturl, body = toJSON(gg.js, auto_unbox headers
f1h<-fr.list$h
#}
#   subset_data <- lmdf.c[lmdf.c$target == targets[i], ]
fs<-fs[order(fs$p,decreasing = T),]
library(lme4)
# pcd<-pc1-pc8
fj10<-join.freqs(data.frame(fj9[,c(1,2)]),data.frame(fj6
  ll$lemmas<-lc$lemma
                    control=glmerControl(optimizer="bob
fj8<-join.freqs(data.frame(fj7[,c(1,2)]),data.frame(fj1[,c(1,3)]))
range<-c("2020-01-01","2020-03-01")

#########################################
 lm2b<-lmer(f.rel~target+in.gp+(1|lemma),lmdf.s)
                      optCtrl = list(trace = 1, maxfu
# tokens<-unlist(strsplit(df_combined$text,"[ .:,;?!\
  #  lim<-limit
  df_norm <- dfa %>%
#      }
}
qx<-qs[2]
# lm1<-lm(f.rel~target*gus.c,lmdf.c)
  stops.j<-c(stops,stops.m,stops.p)
# df_freq <- tokens.r %>%
ptext<-ptdf$text[k]
library(lme4)
#fa1$gu[fa1$WORD%in%gp]<-1
#p.df<-data.frame(gpt=lmdf.c$freq[mg1],human=lmdf.c$freq[mh1],postgpt=lmdf.c$freq[mp1])
#fg1$gu<-fs4$p
ptdf.2<-rbind(tdf1,tdf2,tdf3,tdf4,tdf5,tdf6,tdf7,tdf8,tdf9,tdf10,tdf11)
```